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1.1 Document style and layout (`mainstyle.sty`)

Configures the overall visual appearance, including fonts, margins, sectioning etc.

I use A4 paper with the XITS font. Left and Right margins are 2cm, top and bottom margins are 2.5cm.

- chapter style: Bjornstrup style from `fncychap`.
- section numbering: Section numbering depth is set to 3, including `\subsection` and `\subsubsection`.
- title formatting: all section titles are bold (`\section` is `\Large`, `\subsection` is `\large`) and include a period after the number (e.g., **1.1. Title**).
- Header/Footer Style: `fancyhdr` with 0.4pt rules for both header and footer. Right header displays the current section name (`\rightmark`), left footer displays the current chapter name (`\leftmark`), right footer displays the page number (`\thepage`).
- enumerate list environment uses labels in (1), (2), (3)... format

1.2 Custom colors (`colors.sty`)

“Semantic” palette of colors (named by their purpose). The box packages automatically mix these primary colors with white to generate the lighter backgrounds.

The core semantic colors defined are:

- `ColorDef`: used for definitions (Purple).
- `ColorLemma`: used for lemmas (Blue).
- `ColorThm`: used for theorems and propositions (Green).
- `ColorProof`: used for proofs (Red).
- `ColorGen`: used for generic boxes and text (Grey).
- `ColorCor`: used for corollaries (Bright Green).

Additional colors include `ColorEx` (examples), `ColorNote` (remarks), `ColorWarn` (warnings), and `ColorQues` (questions/exercises).

1.3 Custom macros (macros.sty)

This package defines commands for specific symbols and mathematical sets. There aren't very many, as i am still defining which ones i need (and most of the heavy lifting is done by my nvim config).

Pretty basic:

- `\xor`: XOR symbol ($\oplus$).
- `\textcircled`: improved version that provides better vertical alignment for single characters
- `\N`, `\R`, `\Q`, `\Z`: Standard sets ($\mathbb{N}$, $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$).
- `\mc{X}`: wrapper for `\mathcal{X}`, + `\Aa`, `\Bb` etc for `mathcalA` etc

1.4 Package dependencies (packs.sty)

The `packs.sty` file includes the core packages i use in my notes (mainly maths and formatting packages).

1.5 Title page (title.sty)

Title page is always a work in progress. Right now, it's generic, but i'm still looking for a better one. (the current one is on the last page of this pdf)

All of the boxes in my template are very similar (the designs are different, but the box types are the same). That is because they are iterations of me deciding which style i like best. Right now, I only use “frame boxes”, but at different points i used both “round” and “note” boxes, too.

2.1 Frame Boxes

The boxes defined in `frameboxes.sty` are:

- `defframe`

Def. 1 Title

This is a definition.

- `cdefframe`

Def. 2: Title

This definition allows you to pass a specific semantic color as input. Its counter is shared with `defframe`.

The syntax is: `\begin{cdefframe}{ColorName}{title}`

- `cframe`

Custom Title

This is a custom-colored frame. A color is passed as input like with `cdefframe`, but without any numbering or “Def.”

The syntax is: `\begin{cframe}{ColorName}{title}`

- `lemmaframe`

Lemma 1 The Fundamental Lemma

Is blue, includes “Lemma <counter>”.

- `thmframe`

Thm. 1 **Pythagorean Theorem**

Just like `lemmaframe`, but green and with a different counter.

- `corframe`

Corollary for Thm. 1: **Pythagoras Something**

Corollary! Automatically references the last theorem created and uses `ColorCor`.

Syntax is simply: `\begin{corframe}{title}`

- `propframe`

Prop. 1 **Commutativity**

Just like `thmframe`, but with a different counter.

- `proofframe`

Can be used without title... (useful in nested boxes)

Title Example

But it takes in one input that can be used to override the `notitle` option like so:

`\begin{pframe}[title=Title Example]`

- `gframe`

With title

General-purpose gray note. Uses the `ColorGen` base.

Title can be left empty (`{}`).

The framed shaded environment is also redefined via `tcb` (in light gray):

Usage: `\begin{shaded}[text]\end{shaded}`

and also in an inline version `\shade{[text]}`

2.2 Round Boxes

The boxes defined in `roundboxes.sty` are:

- `defbox`

Def. 1: Title

This is a definition box. Title is preceded by “Def. <counter>”. Uses the `ColorDef` base.

- `lemmabox`

Lemma 1: The Fundamental Lemma

Is blue, includes “Lemma <counter>”. It uses the `ColorLemma` base.

- `thmbox`

Thm. 1: Pythagorean Theorem

Just like `lemmabox`, but green and includes “Thm. <counter>”. Uses the `ColorThm` base.

- `proofbox`

Proof

Can be used without options. Has a fixed default title: “Proof”. Uses the `ColorProof` base.

Alternative Title

It takes one optional argument to override options (e.g., the title).

Syntax: `\begin{proofbox}[title=Alternative Title]`

- `gbox`

With title

General-purpose gray note. It uses the `ColorGen` base.

Syntax: `\begin{gbox}[optional-args]{title}`

Title can be left empty (`{}`).

2.3 Note Boxes

(Hugely inspired by [these](#)).

The boxes defined in `noteboxes.sty` are:

- `defnote`

Def. 1: Title

This is a definition. It uses the `ColorDef` base.

- `lemmanote`

Lemma 1: The Fundamental Lemma

Is blue, includes “Lemma”. It uses the `ColorLemma` base.

■ `thmnote`**Thm. 1: Pythagorean Theorem**

Just like `lemmanote`, but green and with a different counter. It uses the `ColorThm` base.

■ `propnote`**Prop. 1: Commutativity**

Just like `thmnote`, but with a different counter. It uses the `ColorThm` base.

■ `proofnote`**Proof!**

Can be used without options, or with an option to override style (like title). Otherwise has a fixed title: “Proof!”. Uses the `ColorProof` base.

Title Example

Example of title overriding.

Syntax: `\begin{proofnote}[title=Title Example]`

■ `gnote`**With title**

General-purpose gray note. It uses the `ColorGen` base.

Syntax: `\begin{gnote}[optional-args]{title}`

Title can be left empty (`{}`).

■ `pnote`

A simple, red-colored box without a title by default. Uses the `ColorProof` base.

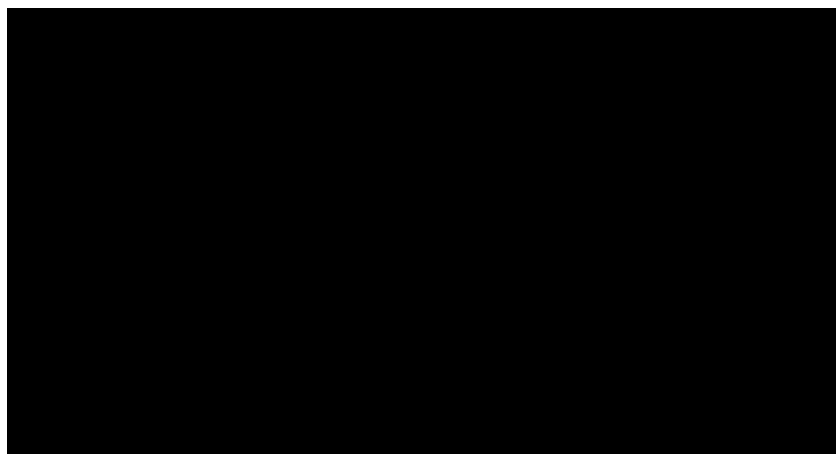
Title Example

It takes one optional argument, which can be used to set a title via `[title=Title Example]`

Sapienza Università di Roma

Course Name

libro del corso:
Book Title, Author Name



Author Name

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